

Welcome to BetterOffice

A faithful, open .docx editor — running a native Rust engine in your browser

BetterOffice is a client-side editor for .docx documents with an unusual architecture: parsing, editing, text shaping, and page layout all run in Rust, compiled to WebAssembly. Your file never leaves your device — there is no upload and no server round trip — yet every page is laid out with the fidelity of a desktop word processor.

The page you are reading is not HTML pretending to be a document. A native layout engine measured every line and painted this page onto a canvas — the same way a desktop word processor would. Try selecting text, changing a heading, or editing a table cell: the engine reflows the page live.

A native engine, compiled to WebAssembly

Opening a file runs the whole pipeline in Rust: the parser reads the OOXML into a typed document model, the text engine shapes every run with real font bytes, the paginator breaks lines into pages, and the result is emitted as a display list that the browser simply replays onto canvas. The DOM is never asked to lay out a document — fidelity does not depend on the browser.

Character formatting

The Rust text shaper keeps character formatting intact: **bold** *italic*, underline, a **colored** **run**, and even a **highlighted run** all survive the round trip.

What runs in Rust

The engine owns the document end to end:

- An OOXML parser and serializer that round-trip .docx files byte-faithfully
- Text shaping and measurement with real font bytes — never the browser
- Pagination: line breaking, tables, floats, headers and footers
- A CRDT editing core, built for multiplayer and agent collaboration

And this is what happens on every keystroke:

1. Your input lands in the resident engine as a single WebAssembly call.
2. The engine edits the document and re-measures only the dirty paragraphs.
3. A binary frame delta repaints just the damaged part of the page.

Architecture at a glance

Where each stage of the pipeline actually runs.

Stage	What happens	Where it runs
Parse & save	OOXML in, OOXML out — the parser and serializer round-trip your file byte-faithfully.	Rust (wasm)
Editing core	Document state, selections, undo, and tracked changes live in a CRDT engine.	Rust (wasm)

Text & layout	Font shaping, line breaking, and pagination — no browser layout involved.	Rust (wasm)
Rendering	The engine emits a display list that is replayed onto page canvases.	Canvas + worker

Table 1. The pipeline — everything before the canvas is native Rust.

Why it matters

Fidelity needs a real engine. Because layout runs in Rust rather than the DOM, the same code can paint this page in every browser today — and in native apps tomorrow.

BetterOffice is open source and maintained by the OpenOOXML project. Explore the code, file an issue, or just keep typing — everything you see is being computed by Rust, right here in your browser.